Quantifying the Environmental Impact of Electric Vehicle Adoption: An Analysis of California's Alternative Energy Vehicle Market

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***Abstract*—This paper examines the environmental impact of the growing popularity of Electric Vehicles (EVs) by quantifying both the Rebound Effect and Substitution Effect in California. Given that California currently has the highest adoption rate of EVs in the United States, it serves as a key indicator for assessing the wider environmental benefits of EVs across the country. We categorize motivations for purchasing EVs into two types: those driven by the Substitution Effect and those attributable to the Rebound Effect. Our analysis suggests that the minimum threshold for replacing Internal Combustion Engine Vehicles (ICEVs) with EVs to achieve real energy savings—according to Jevons' Paradox—is approximately 34%. Furthermore, our review of data from the Alternative Fuels Data Center (AFDC) indicates a downward trend in both electricity and gasoline consumption over the past few years, suggesting that Jevons' Paradox does not apply. Overall, this study not only highlights the environmental implications of EVs but also underscores the importance of considering both the Rebound and Substitution Effects in policies aimed at realizing actual energy savings.**

Keywords—Electric Vehicle, Rebound Effect, Substitution Effect, Jevons' Paradox, Transportation Energy, Greenhouse Gas Emission

I. Introduction

The transportation sector accounts for 28% of all US greenhouse gas (GHG) emissions, 57% of which come from light duty vehicles [1]. Adopting electric vehicles (EVs) can significantly mitigate these emissions. On average, an EV generates only 21.65% of the CO2 equivalent emitted by an internal combustion engine vehicle (ICEV) [2]. EV adoption and resulting population have significantly increased in recent years. The share of EVs among all zero-emission vehicles (ZEVs) surged from 6.56% to 25.06% since 2016, reflecting a marked shift in consumer preference towards fully electric options over hybrids (Fig. 1). Also, drivers are increasingly switching from gasoline vehicles to EVs, motivated by the potential to reduce petroleum use and GHG emissions [3], and attracted by enhanced incentives for cleaner transport [4]. Driving an EV is typically more cost-effective than operating a gasoline-powered vehicle. For instance, the average cost to drive 100 miles in 2020 was $8.81 for an ICEV compared to $4.70 for an EV.5 And the price of electricity tends to be more stable than that of gasoline [5].

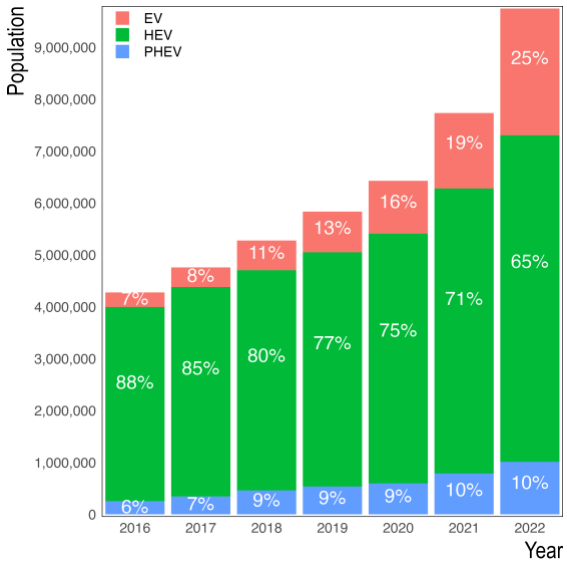

Fig. 1. Population and Proportion of ZEVs in the United States. The data show a steadily increasing trend in the population of ZEVs since 2016. Also, the proportion of EV is monotonically increased and reached 25% in 2022. Source: [6].

Although EV adoption could potentially reduce GHG emissions, it might paradoxically increase emissions due to the Rebound Effect. The Rebound Effect occurs when improvements in energy efficiency lead to increased energy consumption, potentially offsetting or even surpassing the intended benefits [7]. This phenomenon, where technological progress increases energy demand, was first identified by William Stanley Jevons and is known as Jevons' Paradox [8]. Lower costs and emissions per mile for EVs may encourage more driving, which could negate expected gains from reduced emissions. Historically, major technological improvements in efficiency often trigger a Rebound Effect [9]. The lower per-mile costs of EVs may promote more frequent driving [10]. Furthermore, most households purchase EVs as supplementary vehicles rather than replacements for old gasoline cars, leading to a larger overall vehicle population [11]. As more cars are produced and purchased, the negative environmental impact may escalate [11]. It remains uncertain to what extent EV adoption would net a benefit in terms of energy use and GHG emissions, or whether Jevons' Paradox would apply.

This paper develops a break-even point for the Jevons' Paradox concerning vehicle energy consumption in California. A Jevons' Paradox break-even point represents the equilibrium between the Substitution Effect and the Rebound Effect. The Substitution Effect refers to the replacement of

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gas-powered vehicles with EVs, while the Rebound Effect refers to an increase in vehicle fuel use and emissions from increased vehicles miles and number of vehicles on the road.

II. Literature Review

There is substantial literature concerning the Rebound Effect in the context of EV adoption. Our review focuses on major efficiency improvements in the transportation sector that have evidenced the presence of the Rebound Effect, offering examples that support Jevons' Paradox. We also explore the Substitution Effect, which opposes the Rebound Effect by yielding efficiency gains from replacing ICEVs with EVs.

In the US, data from 1979-1994 [12] and a meta-analysis [13] estimate that the long-run Rebound Effect was 15-20% from improved gasoline vehicle efficiency. In Stockholm, 2008 data indicated a 2% lower emissions reduction than expected due to the Rebound Effect [14]. In the UK, from 2001 to 2017, advancements in hybrid and diesel powertrain technology led to a Rebound Effect, negating 4.6% of energy savings [15]. These magnitudes of the Rebound Effect in the US, Stockholm, and the UK indicate that real-world efficiency improvements is achieved with only a minor rebound.

In contrast, from 1990 to 2007, despite advances in fuel efficiency, increased vehicle size, weight, and power led to net increases in fuel consumption and GHG emissions, a classic example of Jevons' Paradox where improved efficiency was offset by increased consumption [8] [16]. Similarly, in Japan from 2010 to 2013, widespread adoption of hybrid electric vehicles (HEVs) failed to yield energy savings due to the Rebound Effect [17]. A subsequent study in Japan found that any potential energy savings from HEVs were offset by increased vehicle sales and driving distances [18].

Varying conclusions from these cases can be attributed to differences in vehicle miles traveled (VMT) among distinct vehicle types from different periods. No Rebound Effect was found among households purchasing more fuel-efficient vehicles in response to fuel economy restrictions, as these vehicles were less desirable for driving (smaller and lower-performing) [19].

To fully understand the Rebound Effect in California with the rise of EVs, it is essential to examine the VMT of these vehicles. A 2008 study in Stockholm found that energy-efficient vehicle (EEV) drivers traveled about 12.2% more than non-EEV owners [14]. Davis reported that the VMT of EVs and Plug-in Hybrid Electric Vehicles (PHEVs) was significantly lower than that of gasoline vehicles in California, possibly due to these vehicles' limited range and the prevalence of multi-vehicle households [20]. However, recent advances in EV technology challenge these earlier findings. The data prior to 2017 predominantly featured short-range EVs, which became less popular over time [21]. More recent data indicate that since 2021, Battery Electricity Vehicles (BEVs) have logged more annual miles than gasoline vehicles [21].

Several studies have examined consumer substitution behavior when transitioning from ICEVs to EVs. A study in Germany found no significant trend for households to purchase additional EVs [22], suggesting that the EV adoption primarily results in a Substitution Effect rather than an increase in vehicle numbers. Notably, the replaced vehicles are generally fuel-efficient, with 12% being HEVs [23].

Recent studies have explored factors driving the Substitution Effect in EVs by examining consumer behavior and purchasing patterns, including government incentives [24] and the charging station accessibility [25]. Many studies have identified the Rebound Effect, focusing on households' driving habits [20] and comparing costs of electricity and gasoline [26]. Despite these insights, there is little up-to-date research that links EV customer behaviors with environmental impacts to comprehensively quantify energy consumption in the transportation sector.

III. Materials and Methods

This study focuses on California as a bellwether for the US, given that its ZEV market share is almost three times the national average (excluding California) as shown in Fig. 2. The proportion of BEVs relative to the overall vehicle population has more than quadrupled during this period. Also, both the proportion of total vehicles and the number of Light-Duty ZEVs have experienced significant growth over the past decade. Notably, in the last five years, the proportion of ZEVs relative to total vehicles has more than tripled, with BEVs showing the most dramatic increase (Fig. 3).

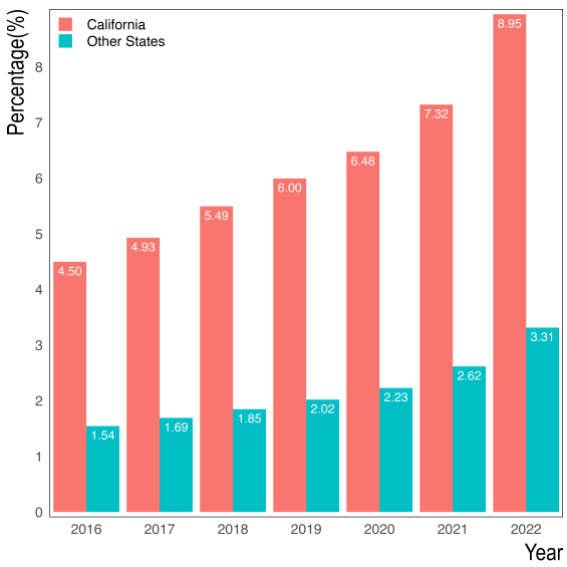

Fig. 2. Percentage of ZEVs Compared to Gasoline Vehicles: California vs. Other States. Both California and other states have experienced an increase in their proportions of zero-emission vehicles since 2016; however, the percentage increase in California is significantly greater than that in other states. Source: [6].

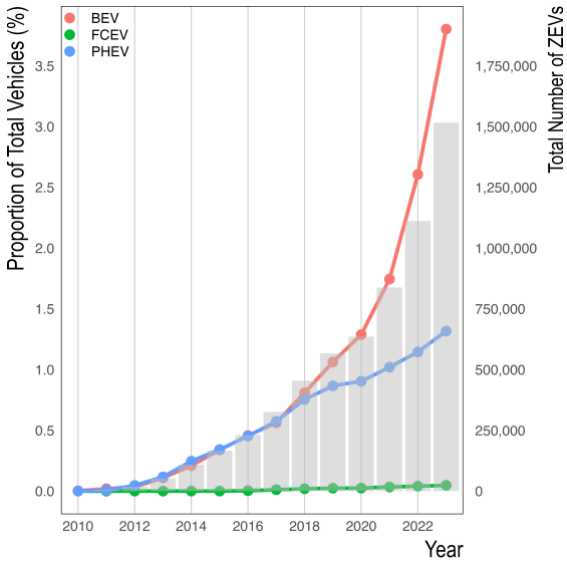

Fig. 3. Proportion and Total Number of ZEVs in California. The graph displays a significant increase in the total number and proportion of BEVs, with a steep rise particularly evident since 2020. In contrast, FCEVs and PHEVs show relatively modest growth over the same period. Source: [27].

We primarily assess environmental impacts in terms of energy consumption, converting the energy consumed by ICEVs and EVs into British Thermal Units (Btu) for straightforward comparisons. To accurately quantify the energy consumption from the rising number of EVs, we categorize EV purchasing behavior into two distinct groups. The first group comprises replacement purchases, driven by the Substitution Effect, where consumers replace conventional vehicles with EVs. The second group includes consumers who buy EVs in addition to their existing vehicles, driven by the Rebound Effect.

Although prior research suggests the Substitution Effect is weaker, often involving the replacement of vehicles with more energy-efficient ones [11] [22] [23], we simplify our calculations using average fuel or electricity consumption. To establish the threshold for Jevons' Paradox in energy consumption, we calculate the Btu of fuel consumed by one ICEV and one EV. This calculation multiplies the annual miles driven by the unit energy consumption—broken down into energy content per gallon of fuel or per kWh of electricity and fuel consumption per mile:

$$\text{Annual Energy Use} = \text{Annual Distance} \cdot \text{Energy Content} \cdot \text{Fuel Efficiency} \quad (1)$$

We found that an EV generates approximately 15.2 million Btu annually, while an ICEV produces around 44.4 million Btu per year (TABLE I). We then assessed the percentage of purchases from the Substitution Effect (savings) and the Rebound Effect (additional costs) to determine net energy savings or costs. By combining this data with the energy costs of the remaining ICEVs, we calculated the total energy consumption. (The process is illustrated in Fig. 4.)

TABLE I. ENERGY METRICS OF EVS AND ICEVS

Parameter	EV		ICEV	
Annual Distance	km/year (mile/year) [21]	20,722 (12,876)	km/year mile/year [21]	14,500 (9,010)
Energy Content	Btu/kWh [28]	3,412	Btu/L Btu/gallon [28]	74698.23 (120,214.29)
Fuel Efficiency	kWh/km (kWh/mile) [29]	0.215 (0.346)	L/km (gallon/mile) [30]	0.0963996 (0.0409836)
Annual Energy Use	Btu/year	15,200,788	Btu/year	44,390,607

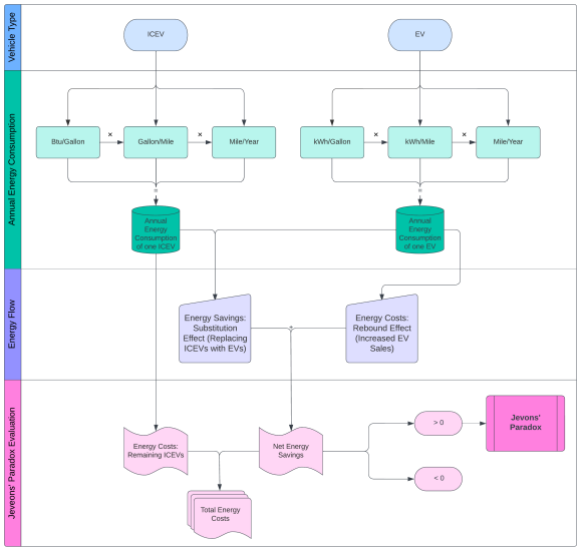

Fig. 4. Quantification of the Rebound Effect: The Model Pipeline. This flowchart outlines the model pipeline for quantifying the rebound effect in the transition from ICEVs to EVs. It explores the impact of vehicle type on energy consumption and efficiency and examines Jevons' Paradox.

To assess net energy savings or costs associated with the rise of EVs, we varied the percentage of purchases attributable to the Substitution Effect and calculated the corresponding Energy Cost using:

$$\text{Energy Cost(p)} = p \cdot \left(EC_{\{EV\}} - EC_{\{ICEV\}}\right) + (1-p) \cdot EC_{\{EV\}} \quad (2)$$

where,

p: Substitution Percentage of ICEVs with EVs
$EC_{\{EV\}}$: Annual Energy Consumption in Btu for One EV
$EC_{\{ICEV\}}$: Annual Energy Consumption in Btu for One ICEV

Equation (2) calculates the overall annual energy consumption by accounting for both the substitution of ICEVs with EVs and the energy usage of the remaining ICEVs. Specifically, it combines the energy savings from substituting percentage p of ICEVs with EVs and the energy consumed by the remaining proportion of EVs. This approach provides an assessment of the net energy impact due to the adoption of EVs in place of ICEVs with a varied substitution rate, p. We then measure the magnitude of the Rebound Effect using:

$$\text{Rebound Effect} = \frac{\left(EC_{\{ICEV\}} - EC_{\{EV\}}\right) + \text{Energy Cost}(p)}{EC_{\{ICEV\}} - EC_{\{EV\}}} \quad (3)$$

Equation (3) quantifies the Rebound Effect by calculating the energy consumption differences between an ICEV and an EV and an additional energy cost dependent on the

substitution percentage, *p*. It then divides this sum by the potential energy savings from replacing an ICEV with an EV, effectively measuring how much the anticipated energy savings are offset by increased usage at different levels of substitution.

IV. Results and Discussion

We found the equilibrium *p* to be approximately 34% (TABLE II). This percentage indicates that energy savings from EV substitutions are exactly offset by additional purchasing behaviors. Specifically, if over 34% of annual EV purchases replace ICEVs and the remaining 66% are additional vehicle acquisitions, the net change in energy costs is zero. This point is also referred to as the threshold of Jevons' Paradox. A replacement proportion below 34% signifies the occurrence of Jevons' Paradox.

Although it is challenging to determine the actual percentage of replacement purchases accurately, the validity of our conclusions is supported by recent trends in energy consumption within the transportation sector. When converting both electricity and gasoline consumption to gasoline gallon equivalent (GGE), we found that the total GGE consumption of either electricity or gasoline shows a downward trend (Fig. 5). This pattern underscores a broader decrease in thermal energy consumption across the transportation sector, regardless of the type of energy. Additionally, there has been a steady increase in vehicle population from 2010 to 2021, while the average GGE consumption per vehicle has shown a marked decrease from over 600 to just over 400 during the same timeframe (Fig. 6). After assessing the environmental impact of EVs in terms of both overall GGE levels and per vehicle, we conclude that Jevons' Paradox does not apply in this context.

TABLE II. Magnitude of the Rebound Effect with Varying Percentages of Substitution Purchases

Substitution Percentage	Energy Costs (Btu)	Rebound Effect Percentage
10%	10,761,727	137%
20%	6,322,666	122%
30%	1,883,605	106%
34%	0	100%
40%	-2,555,455	91%
50%	-6,994,516	76%
60%	-11,433,577	61%
70%	-15,872,637	46%
80%	-20,311,698	30%
90%	-24,750,759	15%
100%	-29,189,819	0%

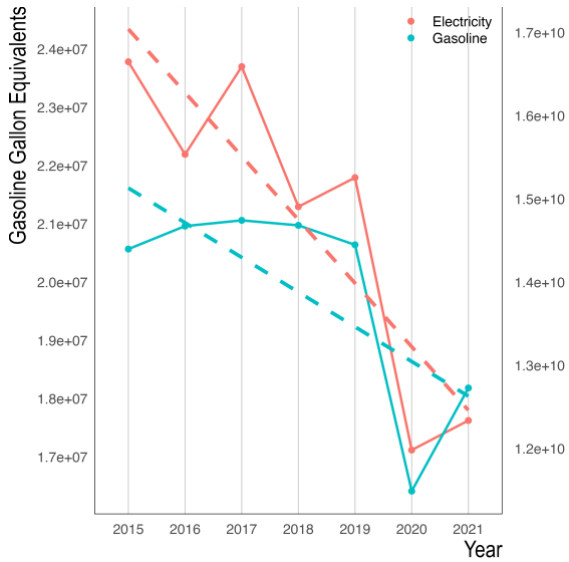

Fig. 5. GGE-Based Energy Consumption from Electricity and Gasoline in California. The graph shows a decline in the GGEs used for both electricity and gasoline from 2015 to 2021, with noticeable volatility in usage patterns throughout the years. Both energy sources experienced sharp declines in recent years, particularly in 2020. Ref: [31].

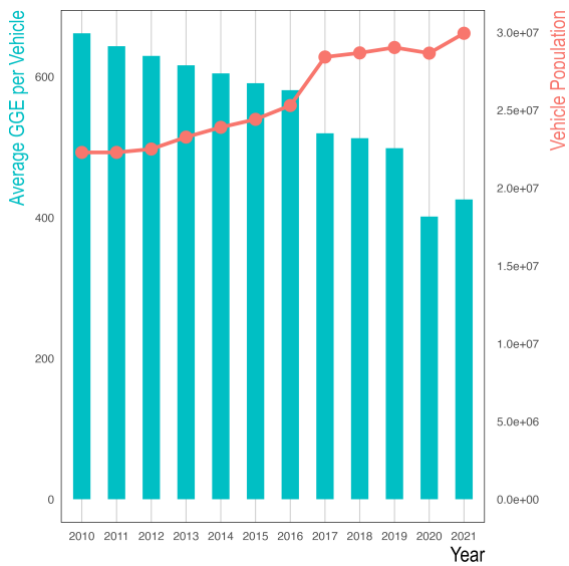

Fig. 6. Average GGE Usage per Vehicle and Total Number of Vehicle Registrations in California. The graph indicates an increasing vehicle population, accompanied by a decrease in average GGE consumption per vehicle. Ref: [27] [31].

V. Conclusion and Future Work

After analyzing driving behaviors and their impact on energy consumption, we established that 34% of substitution purchases are needed to avoid Jevons' Paradox. Our evaluation of fuel consumption—measured in GGE within the transportation sector—indicates that increasing EV adoption does not lead to higher energy consumption. Thus, according to our model and current data, Jevons' Paradox is not evident.

However, limitations, such as the restricted range of EVs in colder areas [32], suggest our California results may not apply to northern US states with similar EV penetration. Given that our analysis is based on average driving distances and fuel consumption for both ICEVs and EVs, the conclusion

should be viewed as a preliminary estimate. Also, the COVID-19 pandemic may have altered driving behaviors during this period. Our study primarily focuses on consumer-side transportation costs, acknowledging the exclusion of significant factors like emissions from car and battery manufacturing, charging infrastructure, and vehicle scrapping. Future research should refine behavioral models and incorporate broader external variables to include these factors.

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